

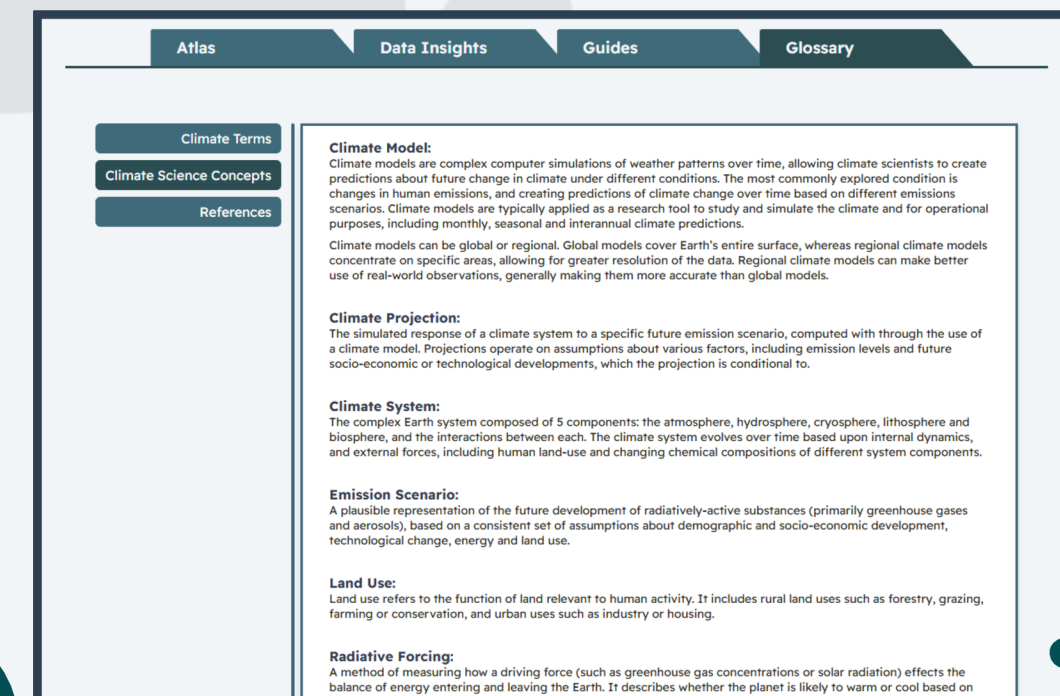
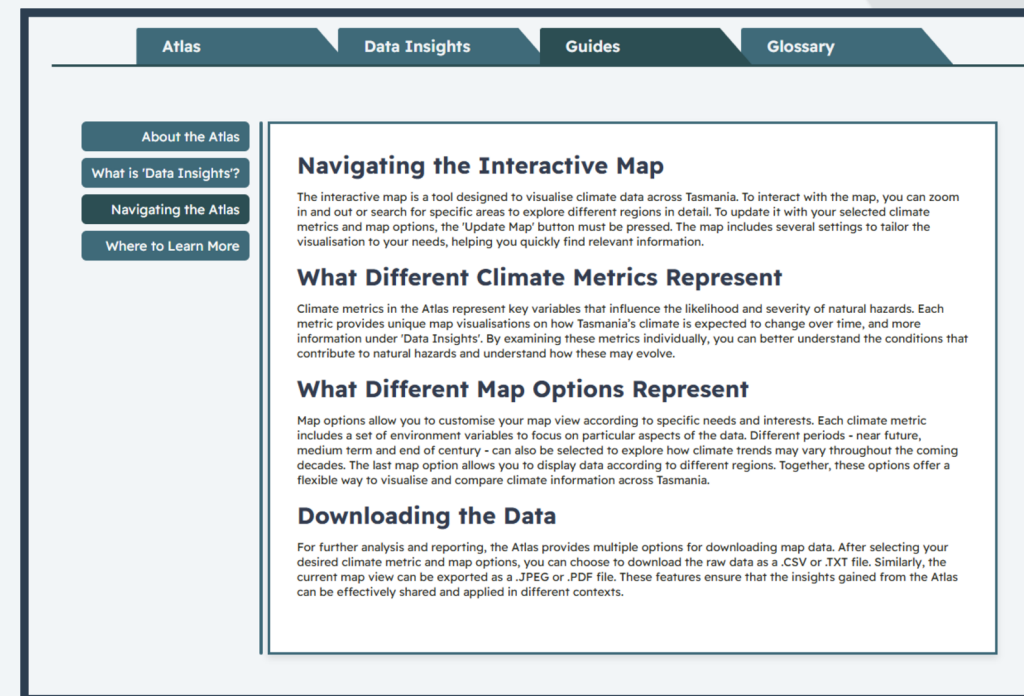
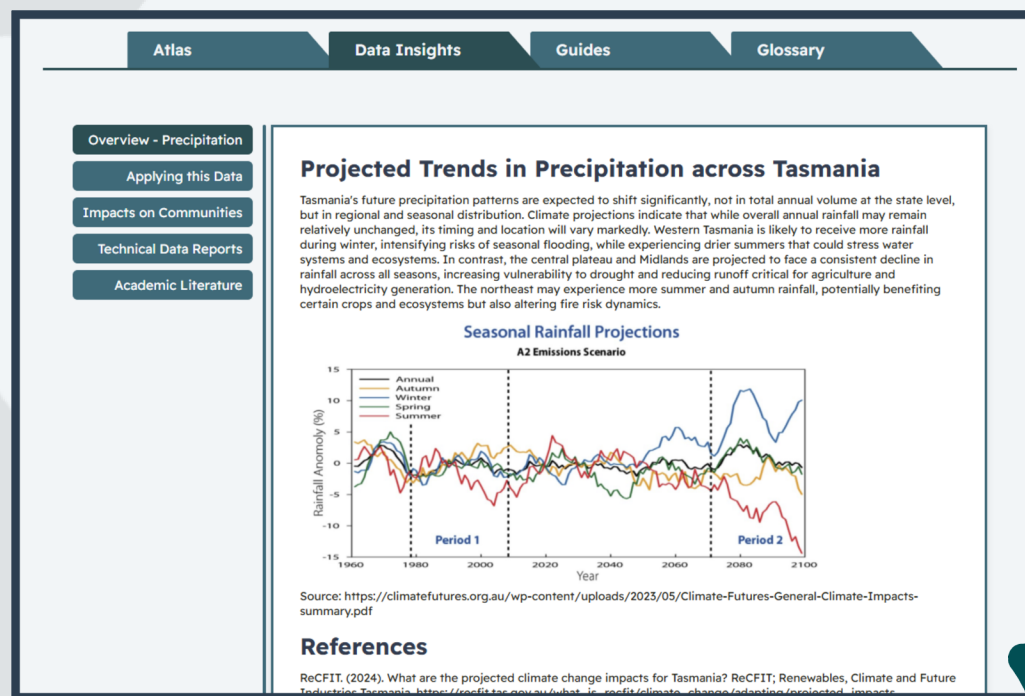
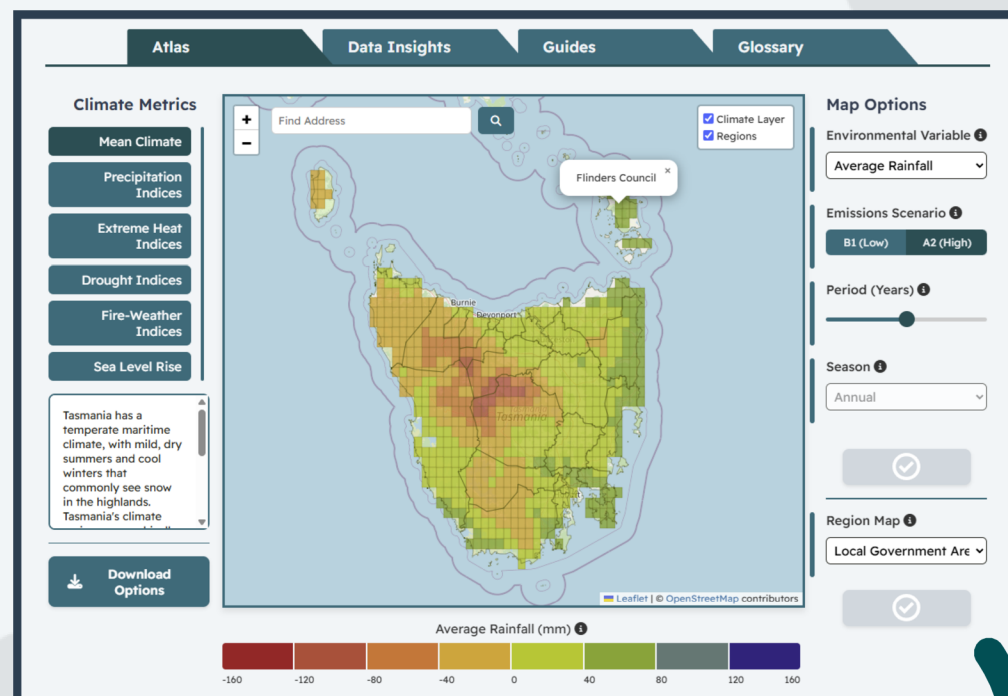
?? About the Atlas ??

The Tasmanian Natural Hazards Atlas is a **climate projection visualisation web platform**. Its goal is to communicate **up-to-date climate research** about Tasmania's **changing climate** to the Tasmanian community, and **assist local communities in preparing for and adapting to climate change**. The Team is incredibly proud to share their work on the Atlas, and are excited to see how it continues to grow after this project.

Tasmanian Natural Hazards Atlas

The Team:

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- Samuel Eleveld



The Map

The Interactive Map, the Data Centre and Heart of the Natural Hazards Atlas.

1. Have a look at the **Map Options**, every combination offers a new dataset to visualise and explore.
2. Whenever you're ready, hit the button: **Update Map**
3. Want to see the climate trends for your community? Set the 'Region Map' to 'Local Government Areas'.
4. Explore change in your selected variable relative to the present:
5. Use the Address Bar to see what's projected for your area.
- 4.5. The Colour Bar below the map describes the values for the colours on the map.

The Data

The Years of Hard Work and Research put in to Prepare us for Our Future.

1. For every Climate Metric on the Atlas Page, there unique **Data Insights** where you can learn about:
 - The Metric itself, it's **Overview** and **Trends** of change
 - Ways to **Apply the Data** the Platform provides
 - How this change is **Impacting Tasmanian Industries and Communities**
- **Technical reports** on how the Platform's data was developed, and the limitations and biases of its development (**not in this version**)
- Links to recent **Academic research** related to the content you're viewing (**not in this version**)
6. You can download the data displayed on the Atlas in two ways and in four formats, using the Download Button in the bottom-left of the Atlas Page.

How to Use

Designed so Anyone and Everyone can Learn from and Apply the Science.

1. **Hint Popups:** ⓘ icons beside Map Options that describe how each variable map to the climate data.
2. **The Guides Page:** Third tab in the navbar offers info on all the key features on the Platform:
3. **The Glossary Page:** A comprehensive Glossary covering Terms and Concepts commonly used in Climate Science discussion.

The Front-End

What are the the Building Blocks?

1. **Vue.js** - Vue is a JavaScript Framework which focuses on modularity. Each major feature is a different component, served dynamically on one page when requested.
2. **Leaflet** - The library used to implement our Interactive Map. It comes with movement, layer control, GeoJSON rendering features, and popups, all used by our platform.
3. **Supporting Content** - All content on the informative pages is in JSON files and served dynamically, making it manageable and scalable.

How Does it Work? - The Back-End

How is the Atlas Converting a Bunch of Tables of Data into... All of this?

1. The Data

Our climate data is stored in 'geo database' (.gdb) format, which is used by most Geographic Information Systems (GIS). The database file contains a files that store geographic information as **features**, where each feature has some coordinates to draw a polygon, and some amount of data values associated with that polygon. This data is produced by a **climate model** - a complex simulation of our climate over a long time period.

2. Processing

This is our platform's **Python** back-end. The front-end makes a request to the back-end's RESTful API Service, which employs the **FastAPI Framework**, with query parameters defining which Map Options are selected. The back-end reads the specified data column from the geodatabase file, processing it into a machine-readable, GGeoJSON format using the **GeoPandas** Library

>/api/update_map>{ }

3. Rendering

The Platform's front-end receives the requested **GeoJSON data**. The **Map.vue** component filters through each feature in the collection to assign the point a colour based on its value. This data is then drawn into a **Leaflet GeoJSON layer**, which is rendered over the map. On mouse-over, the exact value and coordinates of the datapoint is presented in a popup box. Region Maps use the same process as data, being fetched, processed via the back-end API, returned and drawn.